Continuous Glucose Monitoring during Diabetic Pregnancy (GlucoMOMS); a multicentre randomised controlled trial.

Short title: Continuous Glucose Monitoring during Pregnancy

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Conflict of interest

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Abstract

Aim Diabetes is associated with a high risk for adverse pregnancy outcomes. Optimal glycaemic control is fundamental and is traditionally monitored with self-measured glucose profiles and periodical HbA1c measurements. We investigated the effectiveness of additional use of retrospective Continuous Glucose Monitoring (CGM) in diabetic pregnancies.

Material and methods We performed a nation wide multicentre open label randomised controlled trial studying pregnant women with type 1 diabetes, or type 2 diabetes on insulin therapy with gestational age <16 weeks, or insulin treated gestational diabetes with gestational age <30 weeks. Women were randomly allocated (1:1) to intermittent use of retrospective CGM or standard treatment. Women in the CGM group had glycaemic control assessed by CGM for 5-7 days every 6 weeks, while self-monitoring of blood glucose and HbA1c measurements were applied in both groups. Primary outcome was macrosomia, defined as birth weight above the 90th percentile. Secondary outcomes were glycaemic control, maternal and neonatal complications.

Results Between July 2011 and September 2015, we randomized 300 pregnant women with type 1 (n=109), type 2 (n=82), or with gestational diabetes (n=109) to CGM (n=147) or standard treatment (n=153). The incidence of macrosomia was 31.0% in the CGM group and 28.4% in the standard treatment group (relative risk [RR] 1.06, 95%CI 0.83-1.37). HbA1c levels were similar between treatment groups.

Conclusions In diabetic pregnancy, intermittent retrospective CGM use did not reduce the risk of macrosomia. CGM use provides detailed information about glycaemic fluctuations but as a treatment strategy does not translate into improved pregnancy outcome.

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The study is registered at the Dutch Clinical Trial Registry, number NTR2996

http://www.trialregister.nl/trialreg/admin/rctsearch.asp?Term=glucomoms

Introduction

Diabetes during pregnancy is associated with increased risk for complications for both mother and child. Hyperglycaemia is present in approximately 7% of all pregnancies, either pre-existing (diabetes mellitus type 1 (DM1) or type 2 (DM2)) or as a result of gestational diabetes mellitus (GDM).[1] Optimising glycaemic control is a key in prenatal care for diabetic women and is conventionally monitored by means of daily Self Monitoring of Blood Glucose (SMBG) and periodical HbA1c checks. Unfortunately, despite strict target values for glucose and HbA1c, adverse pregnancy outcomes such as pre-eclampsia, caesarean section, high birth weight (macrosomia), shoulder dystocia and perinatal mortality, occur more often in women with diabetic pregnancies as compared to general population.[2;3;4] Macrosomia rates have been reported to be as high as 62.5% of neonates born from women with diabetes.[5;6;7] Macrosomia increases the risk of stillbirth, obstructed labour, birth trauma, and neonatal hypoglycaemia.[8;9] Moreover, Children born macrosomic show even long-term effects such as obesity, and DM2.[10;11;12]

A Continuous Glucose Monitor (CGM) is a modern tool providing detailed insight in daily glucose fluctuations and individual glucose patterns. CGM use is being advised for DM1 patients outside pregnancy under specific conditions.[13] Following a trial showing a reduction of macrosomia, retrospective CGM quickly found its way to

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clinical practice. In this initial trial Murphy et al. randomised 71 women (46 DM1and 25 DM2) to retrospective CGM or SMBG only and reported a reduction of macrosomia (35% versus 60%, OR 0.36, 95% CI 0.13 to 0.98) and improved glycaemic control in terms of a lower HbA1c levels between 32 and 36 weeks of gestation (mean HbA1c 5.6% versus 6.4%).[14] A subsequent larger RCT by Secher et al. compared additional intermittent RT-CGM use with standard treatment in 154 women (123 DM1 and 31 DM2).[15] No improvement was found in macrosomia rate (45% versus 35%, OR 1,51, 95% CI 0.79-2.91) and glycaemic control was similar throughout pregnancy. Both studies were well designed but were small, single or two-centre, and focused on women with pregestational diabetes. Based on a systematic review, including the studies of Murphy and Secher, we previously concluded that there is a need for a larger RCT to clarify the possible benefit of using intermittent CGM during diabetic pregnancy.[16] We therefore conducted the Glucose Monitoring with Sensor (GlucoMOMS) trial, to evaluate the effectiveness of intermittent CGM use in pregnant women with pregestational or gestational diabetes.

Material and Methods

Study design

We performed an open label multicentre pragmatic randomised controlled trial comparing standard care with additional CGM use to standard treatment. The study was performed in 22 hospitals (university, teaching and non-teaching) in the Netherlands and one university hospital in Belgium, collaborating in the Dutch Consortium for Healthcare Evaluation and Research in Obstetrics and Gynaecology.

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The study was approved by the ethics committee of the Academic Medical Centre Amsterdam (reference number MEC AMC 10/322) and by the boards of management of all participating hospitals. The study protocol has been published previously.[17]

Participants

Pregnant women were eligible if they had pre-existing diabetes mellitus and gestational age before 16 weeks or GDM requiring insulin therapy before 30 weeks gestational age. Women with multiple pregnancies, age under 18 or with severe medical or psychological comorbidity were not eligible. Eligible women were identified and counselled by local staff at the outpatient clinic of the obstetric or endocrinology departments or by research nurses. Written informed consent was obtained from all participating women.

Randomisation and masking

Randomisation was performed per centre by a web based computerised program using 1:1 randomisation, stratified according to type of diabetes. Women and clinical staff were not blinded for treatment allocation.

Procedures

Women allocated to CGM were instructed to use the device for 5-7 days every 6 weeks. Glucose profiles were obtained retrospectively directly after each use and were evaluated by the local endocrinologist. Insights were discussed with the patient and dietary or insulin therapy changes were advised accordingly. Hospitals with limited experience with the device at the onset of the trial received personal instructions by the researchers and specialists form Medtronic. In addition, they were

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provided with central support by the trial coordinators during the trial by evaluating monitor results and supplementary patient diary data (information on diet, exercise, SMBG and insulin use) after which an advice was formulated.

We used a retrospective CGM: iPro2™ (Medtronic, Northridge, CA, USA). It determines glucose levels in the extracellular fluid of the abdominal subcutaneous tissue through an electro-chemical reaction and reports values every 10 sec over the range of 2.2 - 22.2 mmol/L. An average value is stored in the monitor every 5 minutes, providing up to 288 glucose values a day. Since the CGM measures interstitial glucose values and exhibits drift, SMBG is required for calibration of the CGM. Patients are unaware of the glucose measurements during CGM use. Afterwards, the readings are uploaded to a web-based program (Carelink™ iPro Therapy Management Software for Diabetes (Medtronic, Northridge, CA, USA)) and presented graphically.

All participating women in both the intervention and the control group performed SMBG (4-8 times/day; at least fasting, after every meal, at bedtime, and preferably also before every meal) and were provided with specific targets: 3.5 - 5.3 mmol/L for fasting state, ≤ 7.8 mmol/L one hour after meals, and ≤ 6.7 mmol/L two hours after meals. HbA1c-levels were measured every 4 weeks throughout pregnancy. Additional obstetric and diabetes care was provided according to local protocols and national guidelines. Medical data were collected with the use of a web-based Case Report Form hosted by an independent data manager.

Study Outcome Measures

The primary outcome was macrosomia, defined as birth weight above the 90th centile for the Dutch population (adjusted for infant sex and gestational age).[18] Secondary

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maternal endpoints were pregnancy induced hypertension (defined as systolic blood pressure ≥ 140 mmHg or diastolic blood pressure ≥ 90 mmHg after 20 weeks of gestation in a previously normotensive woman), pre-eclampsia (defined as pregnancy induced hypertension coinciding with proteinuria of ≥ 300 mg/24 hours), HELLP syndrome (Haemolysis, Elevated Liver enzymes and Low Platelet count), caesarean section, severe hypoglycaemia (defined as requiring the help of another person or hospital admission), and HbA1c levels throughout the course of pregnancy. Secondary neonatal endpoints were birth weight, extremely large for gestational age (birth weight >p95), small for gestational age (SGA) (birth weight <p5) preterm birth (<37 weeks of gestation), neonatal mortality (defined as neonatal death within first 7 days), birth trauma (including shoulder dystocia, clavicle fracture or Erb's palsy), neonatal hypoglycaemia (defined as blood glucose < 2.6 mmol/l and severe as blood glucose <2.0 mmol/l), culture proven sepsis, respiratory distress syndrome, bronchopulmonal dysplasia, intraventricular haemorrhage, and necrotizing enterocolitis (diagnosed according to previously described classifications) and major congenital malformations.[19;20;21]

Statistical analysis

The study was powered to detect a reduction of macrosomia from 45% to 30%, anticipated to outweigh the cost of CGM use. Taking into account a possible 10% protocol violations and drop out rate, we needed a total of 300 women (150 in both arms) (Alpha-error .05, Beta-error .20, one-sided test).

Outcome measurements were compared using chi square test for discrete variables and t-test for continuous variables or Mann Whitney U in case of non-normal distribution, resulting in a relative risk (RR) with accompanying 95% confidence

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interval (CI). Continuous variables are presented as mean with standard deviation (SD) or as median with interquartile range (IQR), as appropriate. Discrete variables are presented as a number and percentage of the total group. The course of HbA1c levels throughout pregnancy was analysed according to LOESS regression method applying a mixed-effects model for longitudinal data with treatment arm, time, and type of diabetes as fixed effects. The primary HbA1c analyses were carried out first using type of diabetes as a covariate in the model, and then separately for the three types of diabetes.

Primary analyses were performed according to intention to treat principle. We also performed a per-protocol analysis limited to women using the CGM at least once every trimester after randomisation. A subgroup analyses was performed based on type of diabetes. All analyses were performed according to a predefined analyses plan.

Data were analysed using SPSS version 22.0.0.0. All analyses were done by two researchers [DV, CN] independently. Mutual agreement was reached in case of discrepancies. HbA1c analyses were done in R Version 3.1.1.

A Data Safety Monitoring Board (DSMB) composed of independent academics [Academic Medical Centre Amsterdam] was assembled before onset of the trial. A planned interim analysis performed on the first 150 participants, disclosed to the DSMB only, gave no reason for premature termination of the trial. The principal investigators [DV, IM] were informed in case of a predefined severe adverse event and reported the event through the web portal *ToetsingOnline* to the primary MEC, within 15 days.

The study was registered at the Dutch Clinical Trial Registry, number NTR2996.

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Role of the funding source

The study was funded by ZonMw, the Dutch organization for Health Research and Development, project number 80-82310-97-11157.The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication. Continuous Glucose Monitors were purchased at a discount price (Medtronic®, Heerlen, the Netherlands). Medtronic® had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between July 2011 and September 2015, we enrolled 300 pregnant women (DM1 (n=109), DM2 (n=82), GDM (n=109)), who were randomly assigned to CGM use (n=147) or standard treatment (n=153). One woman was lost to follow up due to transfer to a non-participating hospital, while four women withdrew consent and five had a miscarriage after randomization. Thus, outcome data were available for 143 women in the CGM group and 147 women in the standard treatment group. The trial profile is represented in figure 1. Baseline characteristics, which were evenly distributed by randomisation, are shown in table 1.

A total of 7 women with DM1 switched to RT CGM, 4 women from the intervention group and 3 from the control group. Furthermore, 44 women from the CGM group refused to (further) comply with the study protocol. Reasons for this included: high burden of the study intervention (n=8), discomfort of the device (n=3), and not

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specified (n=33). Therefore, 95 women from the CGM group (66%) and 144 (98%) women from the standard control group were included in the per-protocol analyses.

A typical CGM read out is given in figure 2.

The primary outcome, incidence of macrosomia occurred in 44 (31.0%) of the births in the intervention group and in 42 (28.4%) of the control group (RR 1.06, 95%CI .83-1.37) (Table 2). For maternal outcomes, we found pregnancy induced hypertension in 19 (13.3%) women in the CGM group and 27 (18.4%)women in the standard treatment group (RR 0.72, 95%CI .42-1.24; p 0.24). There was a statistically significant lower incidence of pre-eclampsia in the CGM group: 5 (3.5%) women vs 27 (18.4%) women in the standard treatment group (RR 0.30, 95%CI 0.12-0.80). HELLP syndrome did not occur in the CGM group but occurred in 4 women in the standard treatment group (RR 0.11, 95% CI 0.01-1.2) The onset of labour and the route of delivery were similar for both groups. There were no maternal deaths.

Mean gestational age at delivery was comparable in both groups. The mean birth weight did not differ statistically significant (3355 grams versus 3423 grams, mean difference 72, 95% CI -226-82). Additionally, no difference was observed in the rate of extremely LGA, or SGA. Other foetal outcomes were comparable in both groups.

One neonatal death occurred. The neonatal death in the control group concerned a neonate of a woman with DM2 in the standard treatment group. Labour was induced at 36 weeks and 5 days of gestation because of severe glycaemic dysregulation. At 9 cm dilation foetal distress occurred and an emergency caesarean section was performed. Three days after birth the macrosomic child died as a result of perinatal asphyxia. In the CGM group one infant died at the 13th day of life. It involved a DM2 pregnancy complicated by premature rupture of membranes followed by premature

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birth at 26 weeks and 5 days. The infant died due to complications of prematurity. Although this woman was randomised to the CGM, she never used CGM. .

HbA1c levels are depicted in figure 2. A LOESS moving average plot showed an initial drop of HbA1c levels during pregnancy and then an increase in the third trimester. A mixed-effect model showed no difference between treatment arms in HbA1c levels throughout the course of pregnancy. The fixed effect of CGM on HbA1c was -0.48mmol/mol (standard error (SE) = 1.12, p =0.66). Subgroup analyses per type of diabetes also did not reveal an association of CGM use with a change in HbA1c levels. The fixed effect per type of diabetes in these analyses was 0.12 mmol/mol (SE = 1.57, p =0.94) for DM1, -0.33 mmol/mol (SE = 2.6, p-value = 0.90) for DM2, and -1.7 mmol/mol (SE = 1.67, p = 0.31) for GDM.

Subgroup analyses for type of diabetes showed no significant difference on rate of macrosomia for DM1, DM2and GDM (table 3) The possible protective effect of CGM on pregnancy induced hypertension, preeclampsia and HELLP seemed restricted to women with type 1 diabetes (Table 4). Results of the per-protocol analysis for macrosomia were 32.6% vs 28.5% (RR 1.15, 95%CI .78-1.69). Maternal and foetal outcomes did not differ significantly when analysed per protocol. Incidence of preeclampsia in the per protocol analysis was 4.2 vs 11.1% (p = 0.06) the number of cases with HELLP syndrome was 0 vs 4 (p = 0.10).

Outcome measures from the compliant (per protocol) women of the CGM group were not significantly different from the non-compliant women of the CGM group.

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Discussion

In this multicentre randomised controlled trial, we showed that intermittent retrospective CGM use did not reduce the risk of macrosomia in women with DM1, DM2 or GDM requiring insulin therapy.

Two previous RCTs on CGM use in pregnancy show contradicting results. Our results do not confirm the positive findings of Murphy et al.[14] They exclusively included women with pregestational diabetes (46 DM1 and 25 DM2) and randomisation was not stratified according to type of diabetes (74% DM1 in the intervention group and 55% in the control group). We included women with pregestational and women with gestational diabetes and stratified randomisation according to type of diabetes. Furthermore, 5 twin pregnancies were included in the study by Murphy, that were all randomised for CGM. Since twin pregnancies are clearly distinctive in foetal growth as well as in perinatal outcomes they were excluded from our study. The incidence of macrosomia in Murphy's CGM group was 35%, which is almost identical to our findings (31%). These numbers are also comparable to other reported incidences of macrosomia as opposed to 60% they found in the control group, which is a strikingly high number. Another noticeable result is that 11% of the CGM group had a baby born SGA (as opposed to 0 in the control group). This might be a reflection of the twin-pregnancies, nonetheless this is an unwanted result and it affects the birth weight analyses. Murphy et al. found a similar number of pre-eclampsia in both groups (2 vs 0 cases).

Our results confirm the results of the other previous trial by Secher et al.[15] They used intermittent RT-CGM and randomisation was stratified according to type of diabetes. They found no difference in pre-eclampsia (8% vs 9%).

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Maternal and foetal outcomes were similar except for a significantly lower incidence of pre-eclampsia in the CGM group, an effect that lost significance in the per protocol analysis and seemed restricted to type 1 diabetes and GDM patients. Glycaemic control as measured by HbA1c levels was similar in both groups. Differences in blood glucose variability and fluctuations are features of glycaemic control that we were not able to investigate in our study but could be a possible explanation. In Murphy's and Secher's trial, there was no indication of less pre-eclampsia in the CGM group. As there was no difference in HbA1c between CGM and standard treatment, there is no obvious explanation for this finding, and chance may have played a role.

Recently, the results of a multicentre randomised controlled trial investigating continuous RT-CGM were reported. The study combined a pregnancy and a planning pregnancy trial and women with DM1 were first enrolled in a run-in phase to determine eligibility for inclusion. Women were then randomised to either continuous RT-CGM use in addition to SMBG (n=108 in the pregnancy trial) or to solely SMBG (n=107). There was a significant difference in large for gestational age infants (defined as birth weight >p90th) with 53% in the CGM group and 69% in the control group. Furthermore, a lower incidence of neonatal hypoglycaemia and less NICU admissions were seen in the CGM group. There was a small difference in HbA1c levels of -0.19%.[22] The positive results of this trial, as opposed to our negative findings, could be explained by the fact that CGM was used continuously. In addition, by using a run-in phase excluding women that encountered technical or motivational difficulties, the study was designed to select a group of woman that was willing to actually use this tool. It must be noted that the rates of macrosomia reported in this trial are strikingly high, especially compared to the incidence of 35% that was achieved by the initial trial by Murphy et al..[14]

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Our study has several strengths. First, our sample size is large, providing enough power to demonstrate a possible effect of CGM use. Secondly, similar to the previous two RCTs on intermittent CGM use, our primary outcome measure is rate of macrosomia. Offering an intervention during pregnancy pursues maternal or foetal benefits and research should be powered on clinically relevant endpoints. Another strength of our study is the inclusion of all types of diabetes. So far, CGM research focused on women with DM1 and DM2. However, the incidence of gestational diabetes is steeply rising and the line between DM2 and GDM is fading. GDM is considered an early stage of DM2 and often women with latent DM2 are screened during pregnancy for the first time, receiving GDM diagnosis and only after pregnancy are shown to actually have DM2. It is therefore an interesting and relevant group to investigate. Our trial is the first RCT on CGM use to include women with insulin treated gestational diabetes, apart from one prospective non-randomized cohort study from China that indicated possible improved outcomes.[23] Furthermore, the external validity of our trial performed in 23 centres is larger than that of the previous single or two centre trials.

Our study also has some limitations. Enrolment took more than four years. Another limitation is the high number of patients that refused continued use of the CGM after the first or second time. A total of 66% of our participants used CGM according to the study protocol. The previous two studies showed comparable numbers, 80% in Murphy's study and 64% in Secher's study used the CGM per protocol. This issue rather reflects an intervention related difficulty in daily practice than a study limitation. It is striking that women who are willing to endure close monitoring of diet, daily multiple needle pricks, frequent hospital visits, while having the health of their baby as a strong motivation, are unable to persist in using the CGM. The nature of this

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intervention strategy (wearing the monitor, extra hospital visits, comprehensive explanation of results and adjustments) comes with a high burden and demands dedication of both patient and professional to continue to use the tool well and translate the information into useful treatment adaptations.[24;25] Finally, we didn't apply blinded CGM in our control group, so we couldn't compare any CGM related outcomes between the two groups.

The results of our trial underline the importance of solid evaluation of an expensive technical tool before wide implementation. Continuous RT-CGM use, as opposed to intermittent use as applied in trials so far, could be a further step in optimizing glucose control and pregnancy outcome, although continuous use is even more demanding for patient and care provider. Furthermore, a small feasibility study on the use of closed loop therapy, combining CGM with an algorithm determining the optimal amount of insulin delivered through a wearable insulin pump during pregnancy, suggests improved glucose control.[26]

As for all interventions that pursue improved glycaemic control it is important to reflect on the clinical benefit, especially during pregnancy where glucose control is an intermediate endpoint and improved pregnancy outcome is the ultimate goal in clinical practice.

CGM registrations provide detailed information on glucose profiles and are a valuable education and research tool. In light of all now available evidence, intermittent CGM use has no place in standard care for pregnant women with diabetes.

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Authors' contributions

AF, BWM, IE were involved in conception and design of the study. DV, JDV, BWM, AF and IE drafted the manuscript. Analysis and interpretation of the data: CN, DV, JDV, BWM, AF, IE. All other authors are local investigators at the participating centers, and participated in the execution of the trial and data collection. All authors edited the manuscript and read and approved the final draft of the manuscript.

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Table 1. Baseline characteristics

	CGM (n=147)	Standard treatment (n=153)
Age (y)	33 (29-36)	32 (29-36)
Gestational age (d)	173 (109-198)	170 (86-197)
Type 1 diabetes Type 2 diabetes Gestational diabetes	50 (34.0) 40 (27.2) 54 (36.7)	56 (36.6) 41 (26.8) 54 (35.8)
Duration of diabetes (y)*	6 (2-13)	5 (3-12)
Preconception HbA1c (mmol/mol)*	51 (42-57)	53 (46-61)
Microvascular complication	11 (7.5)	11 (7.2)
Insulin pump	28 (19.0)	27 (17.6)
Gestational age of diagnosis^ (d)	144 (104-170)	143 (100-176)
Body mass index (kg/m^2)	28 (24-34)	29 (25-34)
Ethnicity White European	95 (64.6)	97 (63.4)
Primiparous	61 (41.5)	59 (38.6)
Folic acid use	69 (46.9)	77 (50.3)

Data presented as median (IQR) or n (%)
*Provided for the cases with type 1 and type 2 diabetes
^ Provided for the cases with gestational diabetes

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Table 2. Outcomes for all women in the intention to treat analysis

	CGM (n=143)	Standard treatment (n=147)	RR, HR (95% CI)
Primary outcome			
Macrosomia, n(%)	44 (31.0)	42 (28.4)	RR 1.08 (0.76-1.54)
Maternal outcomes			
Severe glycaemic dysregulation, n(%)	6 (4.2)	3 (2.0)	RR 2.06 (0.52-8.06)
Pregnancy induced hypertension, n(%)	19 (13.3)	27 (18.4)	RR 0.72 (0.42-1.24)
Preeclampsia, n(%)	5 (3.5)	17 (11.6)	RR 0.30 (0.12-0.80)
HELLP syndrome, n(%)	0	4 (2.7)	RR 0.11 (0.01-1.2)
Onset of labour, n(%)			
Spontaneous	20 (14.0)	24 (16.3)	
Induction	90 (62.9)	87 (59.2)	
Primary caesarean section	33 (23.1)	36 (24.5)	
Route of delivery, n(%)			
Vaginal	71 (65.7)	77 (67.0)	
Instrumental	14 (13.0)	12 (10.4)	
Caesarean section	23 (21.3)	26 (22.6)	
Foetal outcomes			
GA at delivery (d), median (IQR)	266 (259-269)	266 (259-269)	HR 1.09 (0.86-1.37)
Male, n(%)	90 (53.9)	77 (46.1)	RR 1.19 (0.98-1.46)
Birth weight (gr), mean (SD)	3356 (721)	3429 (610)	-72 (-226-82)
Extremely LGA	32 (22.4)	31 (21.1)	RR 1.06 (0.69-1.64)
SGA	4 (2.8)	7 (4.8)	RR 0.59 (0.18-1.96)
Neonatal admission (MC or NICU) , n(%)	29 (22.1)	30 (21.9)	RR 1.01 (0.65-1.59)
Neonatal hypoglycaemia, n(%)	60 (44.8)	58 (41.7)	RR 1.07 (0.82-1.41)
Severe neonatal hypoglycaemia, n(%)	25 (17.5)	25 (17.0)	RR 1.03 (0.62-1.70)
Prematurity, n(%)	31 (21.7)	34 (23.1)	RR 0.94 (0.61-1.44)
Shoulder dystocia, n(%)	7 (4.9)	4 (2.7)	RR 1.79 (0.54-5.97)
Neonatal death, n(%)	1 (0.7)	1 (0.7)	RR 1.02 (0.06-16.17)
Major congenital malformation, n(%)	5 (3.5)	4 (2.7)	RR 1.28 (0.35-4.66)

GA = gestational age (days); LGA = large for gestational age; SGA = small for gestational age; MC = medium care; NICU = neonatal intensive care unit

Table 3. Subgroup analyses for macrosomia per type of diabetes

	CGM	No CGM	RR (95% CI)
Type 1 diabetes, n(%)	22 (44.9)	20 (36.4)	1.24 (0.77-1.97)
Type 2 diabetes, n(%)	11 (27.5)	13 (34.2)	0.80 (0.41-1.57)
Gestational diabetes, n(%)	11 (20.4)	9 (16.7)	1.22 (0.55-2.71)

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Table 4. Subgroup and per protocol analyses on hypertensive disorders

	CGM	Standard treatment	RR (95% CI)	P value
Subgroup type 1 diabetes				
Total number	49	55		
Pregnancy induced hypertension, n(%)	2 (4.1)	9 (16.4)	0.25 (0.06-1.10)	0.04
Preeclampsia, n(%)	1 (2.0)	8 (14.5)	0.14 (0.02-1.08)	0.03
HELLP syndrome, n(%)	0	3 (5.5)		0.10
Subgroup type 2 diabetes				
Total number	40	38		
Pregnancy induced hypertension, n(%)	8 (10.0)	10 (26.3)	0.76 (0.34-1.72)	0.51
Preeclampsia, n(%)	3 (7.5)	2 (5.3)	1.43 (0.25-8.06)	0.69
HELLP syndrome, n(%)	0	1 (2.6)		0.30
Subgroup gestational diabetes				
Total number	54	54		
Pregnancy induced hypertension, n(%)	9 (16.7)	8 (14.8)	1.13 (0.47-2.70)	0.79
Preeclampsia, n(%)	1 (1.9)	7(13.0)	0.14 (0.02-1.12)	0.03
HELLP syndrome, n(%)	0	0		
Per protocol				
Total number	95	144		
Pregnancy induced hypertension, n(%)	14 (14.7)	26 (18.1)	0.82 (0.45-1.48)	0.50
Preeclampsia, n(%)	4 (4.2)	16 (11.1)	0.38 (0.13-1.10)	0.06
HELLP syndrome, n(%)	0	4 (2.8)		0.10

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Figure 1. Trial profile GlucoMOMS

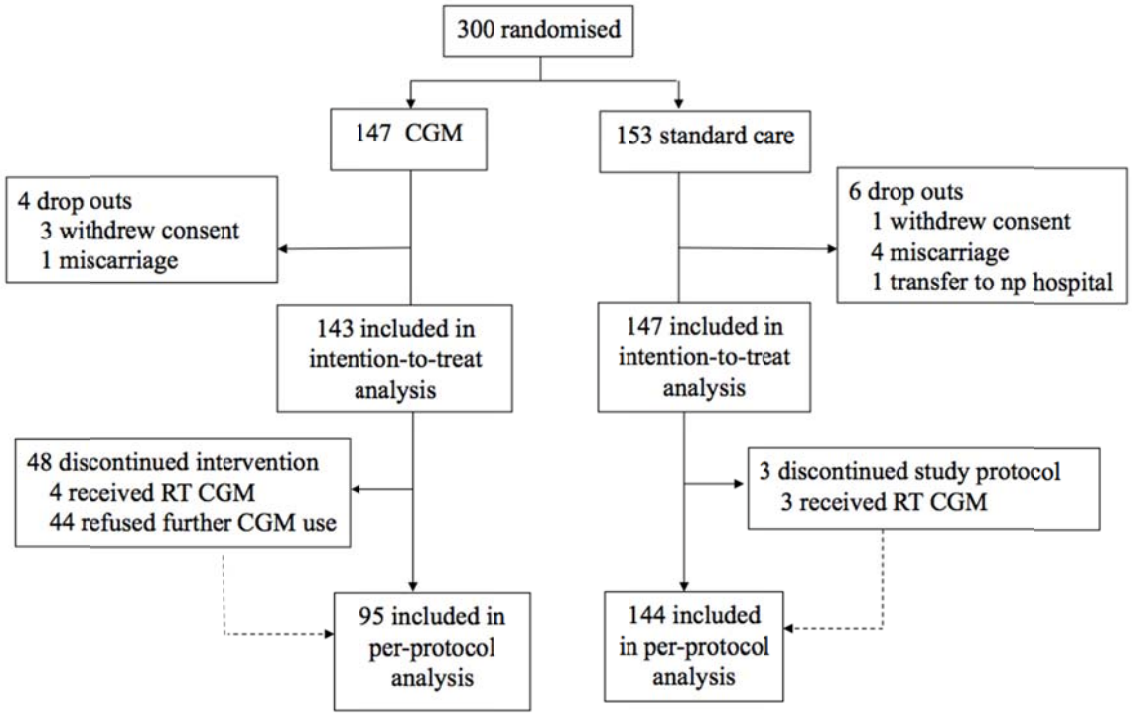

Figure 2. Exemplary CGM readings summary of daily overlay

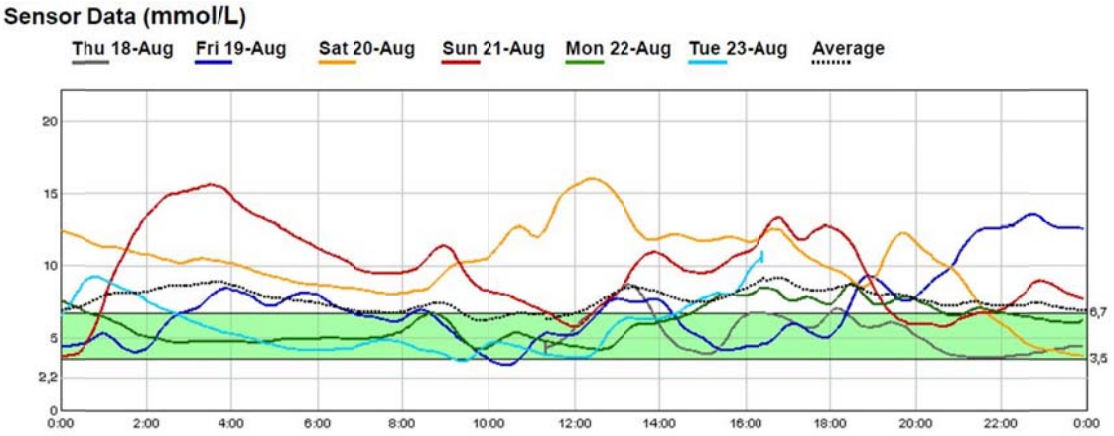

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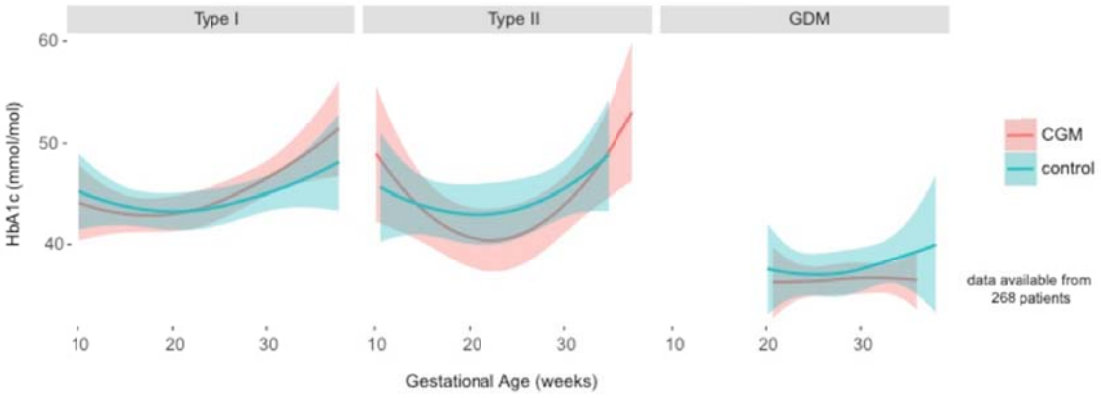
Figure 3. HbA1c measures throughout pregnancy



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